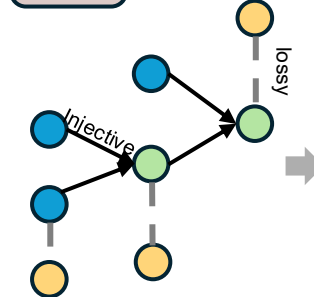


Benchmark Curation

Latent Space

Step 1
Define logic graph

Human

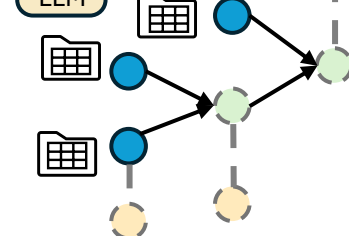


Step 2
Derive minimal set

Algorithm

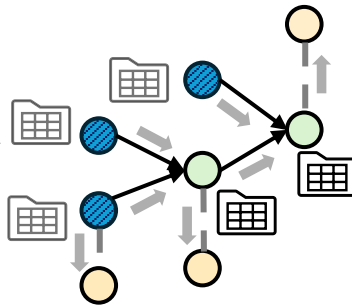
Synthesize data for the minimal set with human instruction

Human



Step 3
Data propagation

Algorithm



● Data ● Image
● Computation Node

Objectives

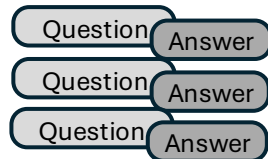
✓ Uniqueness ✓ Consistency ✓ Sufficiency

Surface Space

Step 4 Question Construction

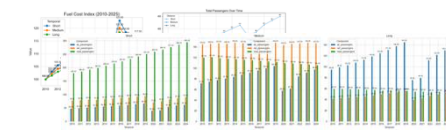
LLM

Synthesize from sample



Step 5 Image Generation

Algorithm



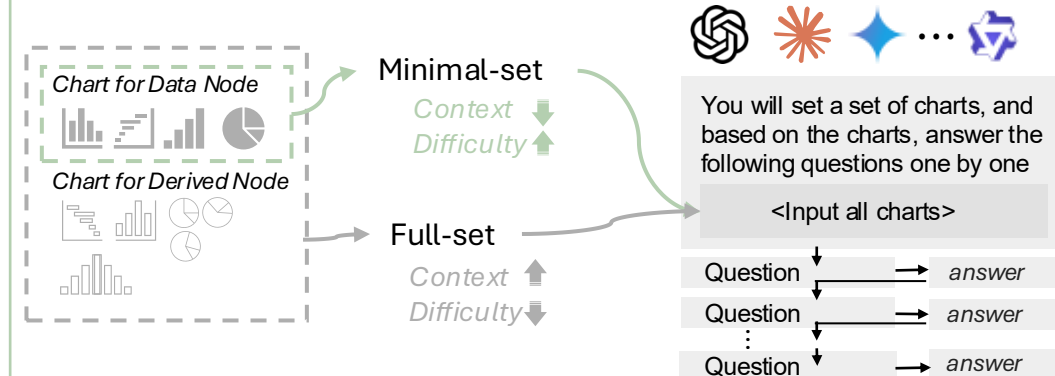
...until images alone formulate a minimal set

Step 6 Human verification

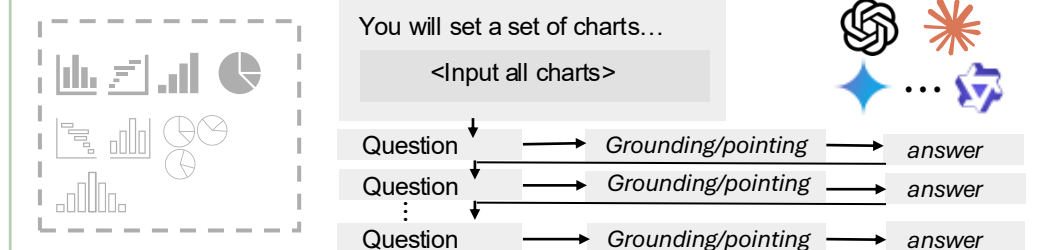
Human

Evaluation Methods

Two Reasoning Pattern



Grounding and Pointing



Three Image Perturbation

